EC015
Chemistry 1
Semester I
Session 2020/2021
2 hours

EC015
Kimia 1
Semester I
Sesi 2020/2021
2 jam

KEMENTERIAN PENDIDIKAN MALAYSIA

BAHAGIAN MATRIKULASI
MATRICULATION DIVISION

PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI
MATRICULATION PROGRAMME EXAMINATION

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU.
DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

Kertas soalan ini mengandungi **10** halaman bercetak.
This question paper consists of **10** *printed pages.*

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INSTRUCTIONS TO CANDIDATE:

This question paper consists of **6** questions.

Answer **all** questions.

Answers must be written in the answer booklet provided.

All steps must be shown clearly. Use a new page for each question.

Maximum marks awarded are shown in brackets at the end of each question or section.

The use of non-programmable scientific calculator is permitted.

***ARAHAN KEPADA CALON*:**

Kertas soalan ini mengandungi **6** *soalan*.

Jawab ***semua*** *soalan.*

Jawapan hendaklah ditulis pada buku jawapan yang disediakan.

Semua langkah kerja hendaklah ditunjukkan dengan jelas. Gunakan muka surat baharu bagi nombor soalan yang berbeza.

Markah maksimum yang diperuntukkan ditunjukkan dalam kurungan pada hujung setiap soalan atau bahagian soalan.

Penggunaan kalkulator saintifik yang tidak boleh diprogramkan dibenarkan.

RELATIVE ATOMIC MASSES OF SELECTED ELEMENTS
JISIM ATOM RELATIF UNSUR-UNSUR TERPILIH

Element	Symbol	Proton Number	Relative Atomic Mass
Aluminium	Al	13	27.0
Silver	Ag	47	107.9
Argon	Ar	18	40.0
Arsenic	As	33	74.9
Gold	Au	79	197.0
Barium	Ba	56	137.3
Beryllium	Be	4	9.0
Bismuth	Bi	83	209.0
Boron	B	5	10.8
Bromine	Br	35	79.9
Iron	Fe	26	55.9
Fluorine	F	9	19.0
Phosphorus	P	15	31.0
Helium	He	2	4.0
Mercury	Hg	80	200.6
Hydrogen	H	1	1.0
Iodine	I	53	126.9
Cadmium	Cd	48	112.4
Potassium	K	19	39.1
Calcium	Ca	20	40.1
Carbon	C	6	12.0
Chlorine	Cl	17	35.5
Cobalt	Co	27	58.9
Krypton	Kr	36	83.8
Chromium	Cr	24	52.0
Copper	Cu	29	63.6
Lithium	Li	3	6.9
Magnesium	Mg	12	24.3
Manganese	Mn	25	54.9
Sodium	Na	11	23.0
Neon	Ne	10	20.2
Nickel	Ni	28	58.7
Nitrogen	N	7	14.0
Oxygen	O	8	16.0
Platinum	Pt	78	195.1
Lead	Pb	82	207.2
Protactinium	Pa	91	231.0
Radium	Ra	88	226.0
Radon	Rn	86	222.0
Rubidium	Rb	37	85.5
Selenium	Se	34	79.0
Cerium	Ce	58	140.1
Cesium	Cs	55	132.9
Silicon	Si	14	28.1
Scandium	Sc	21	45.0
Tin	Sn	50	118.7
Antimony	Sb	51	121.8
Strontium	Sr	38	87.6
Sulphur	S	16	32.1
Uranium	U	92	238.0
Tungsten	W	74	183.9
Zinc	Zn	30	65.4

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Ionisation constant for water at 25 ºC *Hasil darab ion bagi air pada* 25 ºC	K_w	=	1.00×10^{-14} mol^2 dm^{-6}
Molar volume of gases *Isi padu molar gas*	V_m	= =	22.4 dm^3 mol^{-1} at STP 24 dm^3 mol^{-1} at room temperature
Speed of light in a vacuum *Laju cahaya dalam vakum*	c	=	3.0×10^8 m s^{-1}
Specific heat of water *Muatan haba tentu air*		= = =	4.18 kJ kg^{-1} K^{-1} 4.18 J g^{-1} K^{-1} 4.18 J g^{-1} ºC^{-1}
Avogadro's number *Nombor Avogadro*	N_A	=	6.02×10^{23} mol^{-1}
Faraday constant *Pemalar Faraday*	F	=	9.65×10^4 C mol^{-1}
Planck's constant *Pemalar Planck*	h	=	6.6256×10^{-34} J s
Rydberg constant *Pemalar Rydberg*	R_H	= =	1.097×10^7 m^{-1} 2.18×10^{-18} J
Molar of gases constant *Pemalar gas molar*	R	= =	8.314 J K^{-1} mol^{-1} 0.08206 L atm mol^{-1} K^{-1}
Density of water *Ketumpatan air*	ρ	=	1 g cm^{-3}
Freezing point of water *Takat beku air*		=	0.00 ºC
Vapour pressure of water *Tekanan wap air*	P_{water}	=	23.8 torr

UNIT AND CONVERSION FACTOR
UNIT DAN FAKTOR PERTUKARAN

Volume *Isi padu*	1 liter = 1 dm^3 1 mL = 1 cm^3
Energy *Tenaga*	1 J = 1 kg m^2 s^{-2} = 1 N m = 10^7 erg 1 calorie = 4.184 Joule 1 eV = 1.602×10^{-19} J
Pressure *Tekanan*	1 atm = 760 mm Hg = 760 torr = 101.325 kPa = 101325 N m^{-2}
Others *Lain-lain*	1 Faraday(F) = 96500 coulomb 1 Newton(N) = 1 kg m s^{-2}

Answer **all** questions.
*Jawab **semua** soalan.*

1 (a) A species of element **X** has 54 electrons, 56 protons and 81 neutrons. Write the isotopic notation for this species.

Satu spesis unsur **X** *mempunyai* 54 *elektron*, 56 *proton dan* 81 *neutron*. *Tuliskan notasi isotop bagi spesis ini*.

[3 *marks*]
[3 *markah*]

(b) An aqueous solution of 3.58 m potassium bromide, KBr, with a density of 1.12 g cm^{-3} is prepared by adding a certain amount of KBr to 100 g of water. Calculate the molarity and the percentage by mass (% w/w) of the solution.

Suatu larutan akueus kalium bromida, KBr, *berkepekatan* 3.58 m *dengan ketumpatan* 1.12 g cm^{-3} *disediakan dengan menambahkan sejumlah* KBr *kepada* 100 g *air*. *Hitung kemolaran dan peratusan mengikut jisim* (% w/w) *bagi larutan ini*.

[8 *marks*]
[8 *markah*]

(c) An aqueous solution comprising of 0.695 g hydrated copper(II) sulphate, $CuSO_4.nH_2O$, undergoes a complete redox reaction with 0.156 g iron, Fe, to form iron(II) sulphate, $FeSO_4$ and copper, Cu.

Suatu larutan akueus mengandungi 0.695 g *kuprum*(II) *sulfat terhidrat*, $CuSO_4.nH_2O$, *mengalami tindak balas redoks lengkap dengan* 0.156 g *besi*, Fe, *untuk menghasilkan ferum*(II) *sulfat*, $FeSO_4$ *dan kuprum*, Cu.

(i) Write the balanced chemical equation and state the changes in oxidation numbers of Fe and Cu for the redox reaction.

Tulis persamaan kimia berimbang dan nyatakan perubahan nombor pengoksidaan Fe *dan* Cu *bagi tindak balas redoks ini*.

(ii) Calculate the mass of $CuSO_4$ that has completely reacted with the Fe.

Hitung jisim $CuSO_4$ *yang telah bertindak balas lengkap dengan* Fe.

(iii) Calculate the hydration value of water, n.

Hitung nilai penghidratan air, n.

[10 *marks*]
[10 *markah*]

2 (a) Explain why chromium does not obey the Aufbau principle.

Terangkan mengapa kromium tidak mematuhi prinsip Aufbau.

[4 *marks*]
[4 *markah*]

(b) $\mathbf{Y}^{2+}$ ion has 2, 8 and 10 electrons in principal quantum number, n = 1, 2 and 3, respectively. Determine the atomic number of **Y**.

Ion $\mathbf{Y}^{2+}$ *masing-masing mempunyai* 2, 8 *dan* 10 *bilangan elektron bagi nombor kuantum utama,* n = 1, 2 *dan* 3. *Tentukan nombor atom bagi* **Y**.

[2 *marks*]
[2 *markah*]

(c) An electron in hydrogen atom is excited to n = 5 energy level after it absorbs a photon of 2.09×10^{-18} J.

Satu elektron pada atom hidrogen teruja kepada aras tenaga n = 5 *apabila ia menyerap sebanyak* 2.09×10^{-18} J *foton.*

(i) Calculate the energy of this electron.

Hitung tenaga bagi elektron ini.

(ii) Determine the initial energy level of this electron.

Tentukan aras tenaga asal bagi elektron ini.

[4 *marks*]
[4 *markah*]

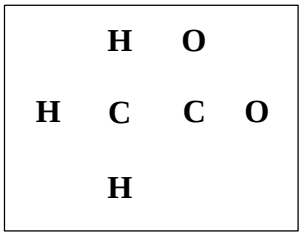
3 (a) Draw the Lewis structure and name the molecular shape of ClF_3 and CF_4. Choose the more polar molecule.

Lukis struktur Lewis dan namakan bentuk molekul bagi ClF_3 *dan* CF_4. *Pilih molekul yang lebih polar*.

[5 *marks*]
[5 *markah*]

(b) The carboxylate ion, $CH_3CO_2^-$, has an arrangement of atoms as shown below:

Ion karbosilat, $CH_3CO_2^-$, *mempunyai susunan atom seperti yang ditunjukkan di bawah*:



(i) Complete the structure of the carboxylate ion by adding bonds and lone pair electrons.

Lengkapkan struktur ion karbosilat dengan menambahkan ikatan dan pasangan elektron pencil.

(ii) Identify the types of hybridization of C atoms in the ion.

Kenalpasti jenis penghibridan atom C *dalam ion tersebut*.

(iii) Show and label the possible intermolecular interaction(s) between CH_3COOH molecules.

Tunjuk dan labelkan interaksi daya antara molekul yang mungkin antara molekul CH_3COOH.

[6 *marks*]
[6 *markah*]

(c) Calcium is an element in Group 2 of the Periodic Table.

Kalsium ialah unsur dalam Kumpulan 2 *Jadual Berkala*.

(i) Sketch the electron sea model diagram.

Lakarkan gambar rajah model lautan elektron.

(ii) Describe the type of bonding formed.

Terangkan jenis ikatan yang terbentuk.

(iii) Explain its conductivity.

Jelaskan kekonduktivitiannya.

[6 *marks*]
[6 *markah*]

4 (a) An amount of 480.0 mL of oxygen gas is collected over water in a jar at 25 ºC and 750.0 mmHg.

Sebanyak 480.0 mL *gas oksigen dikumpul secara sesaran air di dalam bekas pada* 25 ºC *dan* 750.0 mmHg.

(i) If the vapour pressure of water is 23.8 torr at 25 ºC, calculate the partial pressure exerted by the oxygen gas (in mmHg) in the jar.

Jika tekanan wap air ialah 23.8 torr *pada* 25 ºC, *hitung tekanan separa yang dirasai oleh gas oksigen (dalam* mmHg*) dalam bekas tersebut*.

(ii) Calculate the volume and the density of the oxygen gas at STP.

Hitung isipadu dan ketumpatan gas oksigen pada STP.

[7 *marks*]
[7 *markah*]

(b) Octane, C_8H_8, is more volatile than ethanol, CH_3CH_2OH. Explain this statement.

Oktana, C_8H_8, *adalah lebih mudah meruap berbanding etanol*, CH_3CH_2OH. *Jelaskan pernyataan ini*.

[2 *marks*]
[2 *markah*]

5 (a) Steam react with powdered charcoal to produce a combustible gas at 400 °C.

Stim bertindak balas dengan serbuk arang untuk menghasilkan gas yang mudah terbakar pada 400 °C.

$$H_2O(g) + C(s) \rightleftharpoons H_2(g) + CO(g) \qquad \Delta H = +206 \text{ kJ mol}^{-1}$$

When 1.5 atm of steam was initially used, the pressure of the vessel increased to 2.4 atm at equilibrium.

Apabila 1.5 atm *stim digunakan pada awalnya, tekanan dalam bekas meningkat kepada* 2.4 atm *pada keadaan keseimbangan*.

(i) Calculate the partial pressure of all gases in the equilibrium mixture.

Hitung tekanan separa semua gas dalam campuran pada keseimbangan.

(ii) Calculate the equilibrium constant, K_c, at 400 °C.

Hitung pemalar keseimbangan, K_c, *pada* 400 °C.

[7 *marks*]
[7 *markah*]

(b) Explain how the equilibrium constant, K_c, will be affected when the volume of the vessel is reduced to 50%.

Jelaskan apa yang akan terjadi pada pemalar keseimbangan, K_c, *apabila isipadu bekas dikurangkan kepada* 50%.

[2 *marks*]
[2 *markah*]

6 (a) Boric acid, H_3BO_3, is a food addictive that effectively inhibit the growth of yeasts. The pK_a value of boric acid is 9.24.

Asid borik, H_3BO_3*, merupakan bahan tambahan makanan yang berkesan bagi merencat pertumbuhan yis. Nilai* pK_a *bagi asid borik ialah* 9.24.

(i) Write the dissociation equation of H_3BO_3 in water.

Tulis persamaan penguraian H_3BO_3 *dalam air.*

(ii) Calculate the pH of a 0.20 M solution of H_3BO_3.

Hitung nilai pH *bagi larutan* 0.20 M H_3BO_3.

[9 *marks*]
[9 *markah*]

(b) Lead sulphate, $PbSO_4$, is a slightly soluble salt with a solubility product, K_{sp}, of 1.6×10^{-8} at 25 °C.
Plumbum sulfat, $PbSO_4$*, merupakan garam larut sedikit dengan hasil darab keterlarutan,* K_{sp}*, ialah* 1.6×10^{-8} *pada* 25 °C.

(i) Write the K_{sp} expression for $PbSO_4$.

Tulis ungkapan K_{sp} *bagi* $PbSO_4$.

(ii) Calculate the solubility, *s*, of $PbSO_4$ in 0.1 M H_2SO_4 solution.

Hitung keterlarutan, s, bagi $PbSO_4$ *dalam larutan* 0.1 M H_2SO_4.

[5 *marks*]
[5 *markah*]

END OF QUESTION PAPER
KERTAS SOALAN TAMAT